Sensor Fusion for Vehicle State Estimation

(using Kalman Filters)

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Overview

- History and Background
- How It Works
- “Flavors” of Kalman Filters
- Design Example with Real Time Operating System (RTOS)

Terminology

EKF - Extended Kalman Filter

IMU - Inertial Measurement Unit (Accelerometer + Gyroscope)

GNSS/GPS - Global Navigation Satellite System

Linear System - A system of equations composed of constants and linear terms

$$\begin{cases} x_t = x_{t-1} + v_x t \\ y_t = y_{t-1} + v_y t \\ z_t = z_{t-1} + v_z t \end{cases}$$

Non-Linear System - A system of equations composed of at least 1 non-linear term

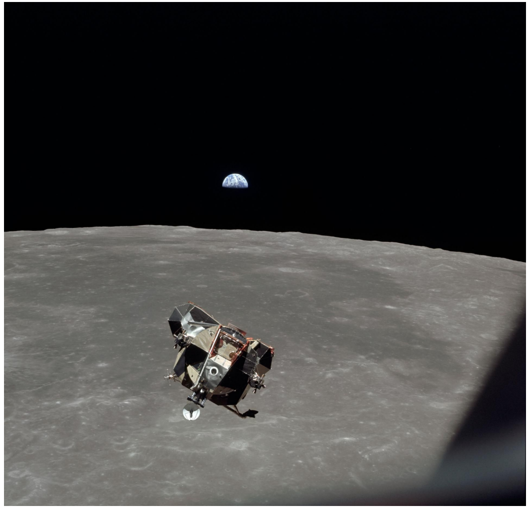
$$\begin{cases} x_t = x_{t-1} + v_x t + \frac{1}{2} a_x t^2 \\ y_t = y_{t-1} + v_y t + \frac{1}{2} a_y t^2 \\ z_t = z_{t-1} + v_z t + \frac{1}{2} a_z t^2 \end{cases}$$

State Vector - A vector representing the current state of the vehicle

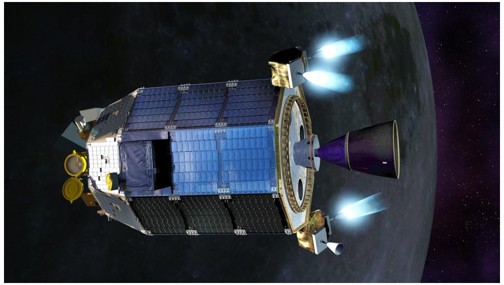
History

- Invented by Dr. Rudolf Kalman, published 1960
- Quickly adopted by researchers at NASA Ames Research Facility in Mountain View, CA for use in the Apollo Guidance Computer
- Modified by NASA engineers to prevent the need for periodic visual observations by the crew for error correction
- Now widely used in various aerospace and robotics applications
    - Self-driving cars
    - Formula 1
    - Manufacturing
    - Spacecraft and Probes

Source



“The Apollo 11 Lunar Module ascent stage, with astronauts Neil A. Armstrong and Edwin E. Aldrin Jr. aboard, is photographed from the Command and Service Modules (CSM) during rendezvous in lunar orbit.” Source



Lunar Atmosphere and Dust Environment Explorer (LADEE) firing it’s thrusters to maintain a safe altitude. Source

Mika Hakkinen’s 1998 championship winning McLaren MP4/13. The 1990s were the beginning of the digital age of Formula 1 Source

Overview — The Kalman Filter

The Kalman Filter, also known as the linear Kalman filter (LKF), is an algorithm that fuses various observations, including noise, into a state estimate that is more accurate than an estimate based on a single measurement

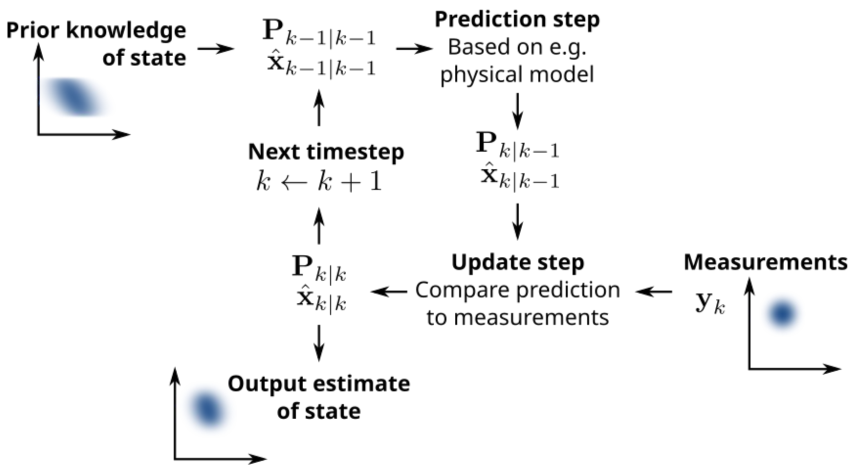
Key Assumptions:

- Both the estimation and observation models are linear
- Noise is modeled as a Gaussian distribution

Kalman Filter - Algorithm

2 Main Components:

- Predict Step
  - Stepping through time according to your system model
  - Control inputs (i.e. steering angle)
- Update Step
  - Sensor Readings (IMU, GNSS)



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Kalman Filter – Predict Step

1. Predict Next State

$$\mu_t = F\mu_{t-1} + Bu_t + w_t$$

- μ_t - State vector estimate at time t
- F - State transition model, applied to μ_{t-1}
- u_t - Control-input vector at time t
- B - Control-input model, applied to u_t
- w_t is process noise at time t
  - $w_t \sim \mathcal{N}(0, Q_t)$

2. Calculate estimate covariance matrix

$$P_t = FP_{t-1}F^T + Q_t$$

- P_t - State covariance matrix at time t
- Q_t - Covariance of w_t

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Update Step – Kalman Gain

Kalman gain is a scalar multiplier between 0 and 1 that represents how much we should trust our prediction vs measurements.

$$K_t = P_{t-1}H_z^T(H_zP_{t-1}H_z^T + R_t)^{-1}$$

- K_t - Kalman Gain at time t
- P_t - State covariance matrix at time t
- H_z - Transition model of observation vector z
- R_t - Covariance of the measurement noise at time t
  - $v_t \sim \mathcal{N}(0, R_t)$

Source

Kalman Filter – Update Step

1. Calculate Kalman Gain
2. Update state estimate

$$\mu_t = \mu_{t-1} + K_t(z - H_z\mu_{t-1})$$

- μ_t - State vector estimate at time t
- K_t - Kalman gain at time t
- z - Observation vector (i.e. GNSS)
- H_z - Transition model of observation vector z

3. Update covariance matrix

$$P_t = (I + KH_z)P_{t-1}$$

- I - Identity matrix of the same dimensions as KH

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Kalman Filter – Pros and Cons

Pros:

- Computationally inexpensive compared to its counterparts
- Ideal estimator for linear systems w/ Gaussian noise

Cons:

- Highly inaccurate in non-linear scenarios, such as cornering at an inconsistent speed.
  - Vehicle Dynamics are non-linear, especially at the limit

The Extended Kalman Filter (EKF)

The extended Kalman filter is a variant of the linear Kalman filter that can account for the non-linearity of a system. It does this by advancing the predicted and observed states using functions rather than transition matrices and linearizing the state.

$$f(\mu_{t-1}, u_t) = \begin{cases} \vec{r}_t = \vec{r}_{t-1} + \vec{v}_{t-1}\Delta t + \frac{1}{2}\vec{a}_u \Delta t^2 \\ \vec{v}_t = \vec{v}_{t-1} + \vec{a}_u \Delta t \\ \dots \end{cases}$$

$h(\mu)$ predicts what the sensor readings should be given state prediction μ

The state and observation transition matrices are still used to calculate the covariance matrix and the Kalman gain, however they are defined as the following jacobians.

$$F_t = \frac{\partial f_j}{\partial \mu_i} \quad H_t = \frac{\partial h_j}{\partial \mu_i}$$

Source

The Extended Kalman Filter – Pros and Cons

Pros:

- Handles non-linearities (i.e. acceleration, tire deformation, cornering at a inconsistent speed)
- Far more stable than LKF for vehicle dynamics

Cons:

- Prone to linearization errors – excess noise
- Can struggle with rotations in certain scenarios
- Can quickly diverge due to poor initialization, absence of observation updates, or linearization errors
- Computationally Expensive compared to LKF
  - Jacobians are calculated and used to calculate transition matrices

Sensor Fusion System Example

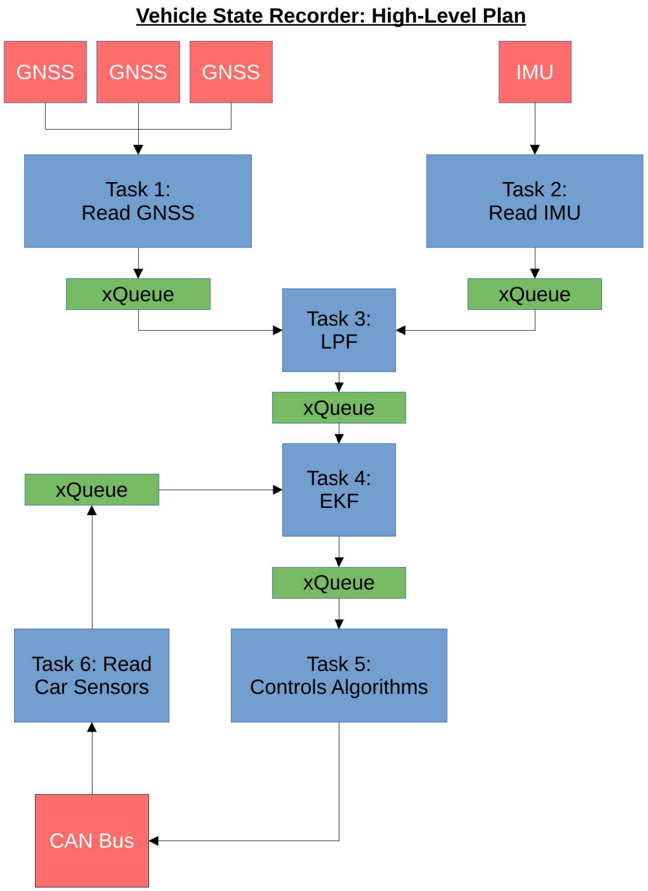
- 3 GPS + 1 IMU Sensor feed sensor readings into an Extended Kalman Filter

Pros:

- Highly integrated — EKF can reduce noise in existing sensor outputs, aiding controls algorithms
- Highly accurate — Can be expanded to use readings from other sensors on the vehicle

Cons:

- Highly Complex — robust documentation required
- Computationally Expensive — affects chip and RAM choices
- Requires quality sensor input — tuning is very important



Questions?

Works Cited

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